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Method for preparing isopropyl alcohol

Abstract

A method of preparing isopropyl alcohol, where a propylene monomer and water are reacted in a reactor to produce a reaction product including isopropyl alcohol, and the reaction product stream is separated into unreacted propylene monomer and isopropyl alcohol with high purity using an absorption column, a first gas purification column, a second gas purification column, and an inert gas removal column.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a US national phase of international application No. PCT/KR2021/018304, filed on Dec. 5, 2021, and claims the benefit of priority to Korean Patent Application No. 10-2021-0070239, filed on May 31, 2021, the entire contents of which are incorporated herein by reference as if fully set forth herein.

TECHNICAL FIELD

(2) The present invention relates to a method of preparing isopropyl alcohol, and more particularly, to a method of separating isopropyl alcohol with high purity from a reaction product of an isopropyl alcohol preparation process and effectively recovering an unreacted material.

BACKGROUND

(3) Isopropyl alcohol (IPA) is used for various purposes such as a cleaning agent in the electronic industry and in the manufacture of semiconductor or liquid crystal display (LCD).

(4) In a process of preparing isopropyl alcohol, for example, propylene and water are used as raw material components, and the propylene and water are reacted to produce isopropyl alcohol.

(5) The reaction product of the isopropyl alcohol preparation process includes isopropyl alcohol, unreacted propylene monomer, and unreacted water. The isopropyl alcohol is separated from the reaction product of the isopropyl alcohol preparation process and recovered, and the unreacted propylene monomer is recovered and reused in the isopropyl alcohol preparation process.

(6) Various processes for separating isopropyl alcohol and an unreacted propylene monomer from the reaction product of the isopropyl alcohol preparation process and recovering and reusing the unreacted propylene monomer have been attempted previously. However, in the conventional isopropyl alcohol preparation process, the recovered unreacted propylene monomer includes isopropyl alcohol in a small amount, and thus, when the propylene monomer is supplied to a reactor and reused, a reverse reaction is caused in the reactor and a propylene conversion rate and selectivity of isopropyl alcohol are reduced. In addition, a propylene monomer is included in a stream supplied to an IPA purification section for purifying isopropyl alcohol, which results in a loss of the propylene monomer.

(7) In addition, inert gas present in a propylene monomer supplied to a reactor is not removed in the conventional isopropyl alcohol preparation process and the inert gas accumulates in the process, and thus, a high-purity propylene monomer is required.

SUMMARY

(8) To solve the problems mentioned above, an objective of the present invention is to provide a method of effectively recovering an unreacted material, and in separating isopropyl alcohol and an unreacted propylene monomer from a reaction product of an isopropyl alcohol preparation process.

(9) That is, an objective of the present invention is to provide a method of preventing isopropyl alcohol from being included in a recovered stream, in recovering an unreacted propylene monomer in a reactor, and preventing unreacted propylene from being included in a stream supplied to an IPA purification section by separating a reaction product stream of a propylene monomer and water using an absorption column, a first gas purification column, a second gas purification column, and an inert gas removal column.

(10) In one exemplary embodiment, a method of preparing isopropyl alcohol includes: supplying a reaction product stream formed by reacting a propylene monomer and water in a reactor to an

absorption column; supplying a lower discharge stream including isopropyl alcohol from the absorption column to a first gas purification column, supplying a part of an upper discharge stream including an unreacted propylene monomer from the absorption column to a second gas purification column, and refluxing the rest of the upper discharge stream to the reactor; supplying an upper discharge stream including the unreacted propylene monomer from the first gas purification column to the second gas purification column and supplying a lower discharge stream including isopropyl alcohol from the first gas purification column to an IPA purification section; supplying an upper discharge stream including the unreacted propylene monomer from the second gas purification column to an inert gas removal column and supplying a lower discharge stream including isopropyl alcohol from the second gas purification column to the IPA purification section; and refluxing an upper discharge stream including the unreacted propylene monomer from the inert gas removal column to the reactor.

(11) According to the method of preparing isopropyl alcohol of the present invention, separation efficiency may be increased by separating a reaction product in an isopropyl alcohol preparation process by a reaction of a propylene monomer and water, using an absorption column, a first gas purification column, a second gas purification column, and an inert gas removal column.

(12) In addition, inclusion of isopropyl alcohol in an unreacted propylene monomer circulating to a reactor by the increased separation efficiency is prevented to promote a forward reaction of an equilibrium reaction in the reactor to increase isopropyl alcohol production, and an outflow of the unreacted propylene monomer into a stream supplied to an IPA purification section is prevented to reduce a loss of the propylene monomer.

(13) In addition, the inert gas removal column is used to remove inert gas, thereby preventing accumulation of the inert gas in the process.

(14) In addition, the supply stages of a stream supplied to each of the second gas purification column and the inert gas removal column are controlled, thereby improving separation efficiency.

(15) In addition, the lower discharge stream from the absorption column is pre-separated using one or more flash drums before supplying the lower discharge stream to the first gas purification column, thereby improving the separation efficiency of the first gas purification column and the second gas purification column and reducing a device size of a rear end column.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIGS. 1 and 2 are process flow diagrams of a method of preparing isopropyl alcohol according to exemplary embodiments of the present invention, respectively.

(2) FIG. 3 is a process flow diagram of a method of preparing isopropyl alcohol according to a comparative example.

DETAILED DESCRIPTION

(3) The terms and words used in the description and claims of the present invention are not to be construed limitedly as having general or dictionary meanings but are to be construed as having meanings and concepts meeting the technical ideas of the present invention, based on a principle that the inventors are able to appropriately define the concepts of terms in order to describe their own inventions in the best mode.

(4) The term “stream” in the present invention may refer to a fluid flow in a process, or may refer to a fluid itself flowing in a pipe. Specifically, the stream may refer to both a fluid flowing in a pipe connecting each device and a fluid flow. In addition, the fluid may refer to a gas or a liquid. Herein, a case in which a solid component is included in the fluid is not excluded.

(5) Hereinafter, the present invention will be described in more detail referring to FIGS. 1 and 2, for better understanding of the present invention.

(6) According to the present invention, a method of preparing isopropyl alcohol is provided. The isopropyl alcohol may be produced by a process of reacting a propylene monomer and water in a vapor phase. Specifically, a feed stream including a propylene monomer and water is supplied to a reactor and a reaction product produced in the reactor may include isopropyl alcohol, an unreacted propylene monomer, and unreacted water. Isopropyl alcohol is separated from the reaction product and recovered, and the unreacted propylene monomer is recovered and reused in the isopropyl alcohol preparation process.

(7) Various processes for separating isopropyl alcohol and an unreacted propylene monomer from the reaction product of the isopropyl alcohol preparation process and recovering and reusing the unreacted propylene monomer have been attempted previously. However, in the conventional process, the recovered unreacted propylene monomer includes isopropyl alcohol in a small amount, and thus, when the propylene monomer is supplied to a reactor and reused, a reverse reaction is caused in the reactor and a propylene conversion rate and selectivity of isopropyl alcohol are reduced and the production of isopropyl alcohol is decreased. In addition, a propylene monomer is included in a stream supplied to an IPA purification section for purifying isopropyl alcohol to cause a loss of the propylene monomer.

(8) In the present invention, to solve the conventional problems described above, a method is provided, in which, when an unreacted propylene monomer is recovered from a reaction product of an isopropyl alcohol preparation process and reused in a reactor, separation efficiency of isopropyl alcohol and a propylene monomer is maximized to prevent isopropyl alcohol from being included in the unreacted propylene monomer to be recovered, thereby promoting a forward reaction of an equilibrium reaction in the reactor to increase production of isopropyl alcohol, and preventing an outflow of the unreacted propylene monomer in a stream supplied to an IPA purification section to reduce a loss of the propylene monomer.

(9) According to an exemplary embodiment of the present invention, a method of preparing isopropyl alcohol includes: supplying a reaction product stream formed by reacting a propylene monomer and water in a reactor to an absorption column **100**; supplying a lower discharge stream including isopropyl alcohol from the absorption column **100** to a first gas purification column **200**, supplying a part of an upper discharge stream including an unreacted propylene monomer from the absorption column to a second gas purification column **300**, and refluxing the rest of the upper discharge stream to the reactor; supplying an upper discharge stream including the unreacted propylene monomer from the first gas purification column **200** to the second gas purification column **300** and supplying a lower discharge stream including isopropyl alcohol from the first gas purification column to an IPA purification section; supplying an upper discharge stream including the unreacted propylene monomer from the second gas purification column **300** to an inert gas removal column **400** and supplying a lower discharge stream including isopropyl alcohol from the second gas purification column to the IPA purification section; and refluxing an upper discharge stream including the unreacted propylene monomer from the inert gas removal column **400** to the reactor

(10) According to an exemplary embodiment of the present invention, a feed stream including a propylene monomer and water may be supplied to the reactor and a gaseous reaction may be performed to prepare a reaction product including isopropyl alcohol. A mole ratio of water to the propylene monomer included in the feed stream may be 0.3 to 0.5, 0.35 to 0.5, or 0.35 to 0.45. By satisfying the mole ratio of water to the propylene monomer in the feed stream supplied to the reactor, a forward reaction of an equilibrium reaction is promoted and the progress of a reverse reaction may be prevented to increase the production of isopropyl alcohol.

(11) According to an exemplary embodiment of the present invention, the reactor may be operated under optimal conditions to efficiently prepare isopropyl alcohol by a gaseous reaction of a propylene monomer and water. For example, the operating pressure of the reactor may be 30 kg/cm² to 50 kg/cm², 35 kg/cm² to 50 kg/cm²,

or 35 kg/cm.sup.2.Math.g to 45 kg/cm.sup.2.Math.g, and the operating temperature may be 180° C. to 220° C., 185° C. to 220° C., or 190° C. to 215° C. By operating the reactor in these pressure and temperature ranges, isopropyl alcohol may be effectively produced by a gaseous reaction using the propylene monomer and water.

(12) According to an exemplary embodiment of the present invention, the reaction product produced by the gaseous reaction of the propylene monomer and water in the reactor is gaseous, and may include isopropyl alcohol, unreacted propylene monomer, and unreacted water. The gaseous reaction product discharged from the reactor may be condensed into a liquid reaction product using one or more heat exchangers and then supplied to an absorption column **100**. The temperature of the reaction product supplied to the absorption column **100** is, for example, 105° C. to 150° C., 110° C. to 140° C., or 115° C. to 140° C.

(13) According to an exemplary embodiment of the present invention, the gaseous reaction product discharged from the reactor may be heat-exchanged with a feed stream supplied from one or more heat exchangers to the reactor. The feed stream passing through the one or more heat exchangers may be heated to the temperature of, for example, 170° C. to 210° C., 180° C. to 200° C., or 185° C. to 195° C.

(14) As such, by heat-exchanging the gaseous reaction product stream discharged from the reactor with the feed stream supplied to the reactor, the gaseous reaction product discharged from the reactor may be condensed into a liquid reaction product and also the feed stream may be supplied to the reactor after being pre-heated. Thus, energy for heating the feed stream to an optimal temperature before supplying the feed stream to the reactor may be reduced to efficiently preparing isopropyl alcohol, and coolant costs required for condensation to a temperature for increasing separation efficiency by supplying the gaseous reaction product to the absorption column **100** may be reduced.

(15) According to an exemplary embodiment of the present invention, the gaseous reaction product passes through one or more heat exchangers to be partly condensed to a liquid reaction product and the rest thereof may be present as the gaseous reaction product. As an example, the gaseous reaction product discharged from the reactor may pass through one or more heat exchangers to be separated into a gaseous first reaction product stream and a liquid second reaction product stream. Here, the first reaction product stream and the second reaction product stream may be separated and discharged through a separate pipe formed in a rear end heat exchanger, or separated through a gas-liquid separation device installed at a rear end of the heat exchanger.

(16) According to an exemplary embodiment of the present invention, a ratio of the flow rate of the first reaction product stream to the flow rate of the second reaction product stream may be 5 to 11, 6 to 10, or 7 to 9. As described above, in a process of heat-exchanging the gaseous reaction product stream discharged from the reactor with the feed stream in one or more heat exchangers, the gaseous reaction product stream discharged from the reactor is cooled to a temperature of 105° C. to 150° C., thereby controlling the ratio of the flow rate of the first reaction product stream to the flow rate of the second reaction product stream to 5 to 11. Herein, the “flow rate” may be measured in units of weight per unit hour. As a specific example, the unit of the flow rate may be ton/hr.

(17) According to an exemplary embodiment of the present invention, the first reaction product stream may be further condensed using a cooler and supplied to the absorption column **100** as a gas-liquid mixed phase in which a gas phase and a liquid phase are mixed. Here, the first reaction product stream passes through one or more heat exchangers and is in a state of being cooled as described above, and the coolant used in the cooler may be replaced with low-priced cooling water and the usage of the cooling water may be minimized.

(18) According to an exemplary embodiment of the present invention, the gaseous first reaction product stream may include 85 wt % to 95 wt % of the unreacted propylene monomer, 4 wt % to 8 wt % of isopropyl alcohol, and 1 wt % to 5 wt % of water. Specifically, the first reaction product stream includes a high content of the unreacted propylene monomer and a low content of isopropyl

alcohol and water.

(19) In addition, the liquid second reaction product stream may include 1 wt % to 10 wt % of the unreacted propylene monomer, 5 wt % to 15 wt % of isopropyl alcohol, and 80 wt % to 90 wt % of water. Specifically, the second reaction product stream may have a low content of the unreacted propylene monomer and a high content of water. The content of isopropyl alcohol included in the second reaction product stream may be higher than the content of isopropyl alcohol included in the first reaction product stream.

(20) According to an exemplary embodiment of the present invention, the first reaction product stream and a liquid second reaction product stream in a gas-liquid mixed phase may be supplied to a lower side of the absorption column **100** as individual streams or a mixed stream.

(21) According to an exemplary embodiment of the present invention, the total number of stages of the absorption column **100** may be 10 to 30, 15 to 30, or 15 to 25. For example, when the total number of stages of the absorption column **100** is 10, the first reaction product and the second reaction product may be supplied to the 10th stage of the absorption column **100**.

(22) According to an exemplary embodiment of the present invention, the reaction product is supplied to the absorption column **100**, and a lower discharge stream including isopropyl alcohol and an upper discharge stream including the unreacted propylene monomer may be separated in the absorption column **100**. Here, the upper discharge stream from the absorption column **100** may further include inert gas together with the unreacted propylene monomer, and the lower discharge stream may further include unreacted water together with isopropyl alcohol. Specifically, the reaction product may be supplied to a lower end of the absorption column **100**, isopropyl alcohol included in the reaction product is dissolved using a solvent supplied to the absorption column **100** to be separated into a lower portion of the absorption column **100**, and the stream including the unreacted propylene monomer may be separated into an upper portion.

(23) The solvent used in the absorption column **100** may be, for example, water. By using water which is a component used in the reaction as the solvent, a separate device for separating the solvent in the rear end is not required.

(24) The flow rate of the solvent supplied to the absorption column **100** may be 20% to 50%, 20% to 45%, or 25% to 40% of the total flow rate of the reaction product supplied to the absorption column **100**. The solvent is supplied at the flow rate within these ranges to improve the absorbency of isopropyl alcohol included in the reaction product supplied to the absorption column **100** while preventing an excessive increase of energy costs for recovering the solvent at the rear end.

(25) According to an exemplary embodiment of the present invention, the operating pressure of the absorption column **100** may be 20 kg/cm².Math.g to 40 kg/cm².Math.g, 25 kg/cm².Math.g to 40 kg/cm².Math.g, or 25 kg/cm².Math.g to 35 kg/cm².Math.g, and the operating temperature thereof may be 80° C. to 110° C., 90° C. to 110° C., or 90° C. to 100° C. By operating the absorption column **100** at these pressure and temperature ranges, the upper discharge stream including the unreacted propylene monomer and the lower discharge stream including isopropyl alcohol may be effectively separated.

(26) According to an exemplary embodiment of the present invention, the lower discharge stream from the absorption column **100** may include a small amount of the unreacted propylene monomer in addition to isopropyl alcohol and unreacted water. For example, the content of the unreacted propylene monomer included in the lower discharge stream from the absorption column **100** may be 5 wt % or less or 3 wt % to 5 wt %. Thus, all or a part of the amount of the lower discharge stream including isopropyl alcohol from the absorption column **100** is supplied to a first gas purification column **200** to separate the unreacted propylene monomer included in the lower discharge stream from the absorption column **100** as an upper discharge stream, and a lower discharge stream including isopropyl alcohol and unreacted water may be supplied to an IPA purification section.

(27) According to an exemplary embodiment of the present invention, the lower discharge stream

from the absorption column **100** may pass through one or more, one to three, or one or two flash drums before being supplied to the first gas purification column **200**. For example, a part of the lower discharge stream from the absorption column **100** may be supplied to the first gas purification column **200**, using a first flash drum **500** or the first flash drum **500** and a second flash drum **510**.

(28) As an example, when one flash drum is used, the lower discharge stream from the absorption column may be supplied to the first flash drum, the unreacted propylene monomer included in the lower discharge stream from the absorption column may be separated as an upper discharge stream from the first flash drum and supplied to the first gas purification column, and a lower discharge stream from the first flash drum may be supplied to the IPA purification section.

(29) Before the lower discharge stream from the absorption column is supplied to the first flash drum, the stream may be depressurized using a first valve and then supplied to the flash drum, thereby improving the separation efficiency of a gas phase and a liquid phase in the first flash drum **500**.

(30) The lower discharge stream from the absorption column is pre-separated through the first flash drum, and a part of the stream including the unreacted propylene monomer and inert gas is supplied to the first gas purification column and another part of the stream including isopropyl alcohol and unreacted water may be supplied to the IPA purification section.

(31) In addition, by controlling the composition of the stream supplied to the second gas purification column to improve separation efficiency, the outflow of isopropyl alcohol into an upper discharge stream from the second gas purification column may be prevented. For example, the contents of isopropyl alcohol and water in the stream supplied to the second gas purification column may be decreased to 5 wt % or less to prevent the outflow of isopropyl alcohol into an upper discharge stream from the second gas purification column.

(32) As another example, the lower discharge stream from the absorption column **100** may be supplied to the first flash drum **500**, the unreacted propylene monomer included in the lower discharge stream from the absorption column **100** may be separated as the upper discharge stream and supplied to a second gas purification column **300**, and the lower discharge stream may be supplied to the second flash drum **510**.

(33) Before supplying the lower discharge stream from the absorption column **100** to the first flash drum **500**, the stream may be depressurized using a first valve **600** and then supplied to the first flash drum **500**. In addition, before the lower discharge stream from the first flash drum **500** is supplied to the second flash drum **510**, the stream may be depressurized using a second valve **610** and then supplied to the second flash drum **510**. Thus, separation efficiency of a gas phase and a liquid phase in each of the first flash drum **500** and the second flash drum **510** may be improved.

(34) The lower discharge stream including isopropyl alcohol and unreacted water from the second flash drum **510** may be supplied to the IPA purification section, and the upper discharge stream including the remaining components may be supplied to the first gas purification column **200**.

(35) The lower discharge stream from the absorption column **100** may be pre-separated using the first flash drum **500** and the second flash drum **510**, a part of the stream including the unreacted propylene monomer and inert gas may be supplied to the second gas purification column **300**, and another part of the stream including isopropyl alcohol and unreacted water may be supplied to the IPA purification section. In addition, the contents of isopropyl alcohol and water in the stream supplied to the second gas purification column **300** are decreased to 5 wt % or less to increase the separation efficiency of the second gas purification column **300** to prevent isopropyl alcohol from being included in the stream supplied to the inert gas removal column **400**, thereby recovering the unreacted propylene monomer with high purity which does not include isopropyl alcohol in the inert gas removal column **400**, and refluxing the unreacted propylene monomer to the reactor and reusing it.

(36) As such, when the lower discharge stream from the absorption column **100** is pre-separated

using one or more flash drums, the lower discharge stream from the absorption column **100** is depressurized and a space for separating a gas phase and a liquid phase before supplying the stream to a rear end column is provided, thereby preventing a problem of inefficiently increasing a size of the column for gas-liquid separation in the rear end column, and improving the separation efficiency of the second gas purification column **300**.

(37) According to an exemplary embodiment of the present invention, a flow rate ratio between the flow rate of the upper discharge stream from the first flash drum **500** and the flow rate of the upper discharge stream from the second flash drum **510** may be 1:0.4 to 1:0.8, 1:0.5 to 1:0.8, or 1:0.5 to 1:0.7. By controlling the flow rate ratio of the upper discharge stream from the first flash drum **500** and the upper discharge stream from the second flash drum **510** within the above ranges a pre-separation effect is improved and also the components and the flow rate of the stream supplied to the first gas purification column **200** and the second gas purification column **300** may be controlled to reduce the device size of the column and improve separation efficiency rate.

(38) According to an exemplary embodiment of the present invention, the operating temperature in the first flash drum **500** may be 80° C. to 100° C., 85° C. to 100° C., or 85° C. to 95° C. In addition, the operating pressure in the first flash drum **500** may be 10 kg/cm.sup.2.Math.g to 25 kg/cm.sup.2.Math.g, 15 kg/cm.sup.2.Math.g to 25 kg/cm.sup.2.Math.g, or 15 kg/cm.sup.2.Math.g to 20 kg/cm.sup.2.Math.g. When the operating conditions in the first flash drum **500** are controlled to be in these ranges, the separation efficiency may be improved.

(39) The operating temperature in the second flash drum **510** may be 80° C. to 95° C., 85° C. to 95° C., or 85° C. to 90° C. In addition, the operating pressure in the second flash drum **510** may be 0 kg/cm.sup.2.Math.g to 2 kg/cm.sup.2.Math.g, 0 kg/cm.sup.2.Math.g to 1 kg/cm.sup.2.Math.g, or 0 kg/cm.sup.2.Math.g to 0.5 kg/cm.sup.2.Math.g. When the operating conditions in the second flash drum **510** are controlled to be in these ranges, the separation efficiency may be improved.

(40) According to an exemplary embodiment of the present invention, the upper discharge stream including the unreacted propylene monomer and the lower discharge stream including isopropyl alcohol may be separated in the first gas purification column **200**. Here, the upper discharge stream from the first gas purification column **200** may further include inert gas together with the unreacted propylene monomer, and the lower discharge stream may further include unreacted water together with isopropyl alcohol. Specifically, the upper discharge stream including the unreacted propylene monomer in the first gas purification column **200** may be supplied to the second gas purification column **300** and the lower discharge stream including isopropyl alcohol may be supplied to the IPA purification section.

(41) The operating pressure of the first gas purification column **200** may be 0 kg/cm.sup.2.Math.g to 2 kg/cm.sup.2.Math.g, 0 kg/cm.sup.2.Math.g to 1 kg/cm.sup.2.Math.g, or 0 kg/cm.sup.2.Math.g to 0.5 kg/cm.sup.2.Math.g, and the operating temperature thereof may be 40° C. to 80° C., 50° C. to 80° C., or 50° C. to 70° C. By operating the first gas purification column **200** in these temperature and pressure ranges, the propylene monomer may be separated with high purity into the upper discharge stream.

(42) According to an exemplary embodiment of the present invention, the stream supplied to the second gas purification column **300** may be supplied as individual streams, or may form a mixed stream and be supplied to the second gas purification column **300**. For example, a part of the upper discharge stream from the absorption column **100** and the upper discharge stream from the first gas purification column **200** may form a mixed stream and be supplied to the second gas purification column **300**, and if necessary, a part of the upper discharge stream from the absorption column **100**, the upper discharge stream from the first gas purification column **200**, and the upper discharge stream from the first flash drum **500** may form a mixed stream and be supplied to the second gas purification column **300**.

(43) According to an exemplary embodiment of the present invention, in the second gas purification column **300**, the upper discharge stream including the unreacted propylene monomer

may be separated from the supplied stream and supplied to the inert gas removal column **400**, and the lower discharge stream including isopropyl alcohol and water may be separated and supplied to the IPA purification section. Here, the entire amounts of isopropyl alcohol and unreacted water in the stream supplied to the second gas purification column **300** may be separated as a lower discharge stream. In this case, since the upper discharge stream from the second gas purification column **300** supplied to the inert gas removal column **400** does not include isopropyl alcohol and water, no isopropyl alcohol may be included in the unreacted propylene monomer recovered from the inert gas removal column **400** and refluxed to the reactor.

(44) According to an exemplary embodiment of the present invention, the operating pressure of the second gas purification column **300** may be 12 kg/cm² to 20 kg/cm², 15 kg/cm² to 20 kg/cm², or 15 kg/cm² to 17 kg/cm², and the operating temperature thereof may be 30° C. to 170° C., 40° C. to 170° C., or 40° C. to 160° C. By operating the second gas purification column **300** at these pressure and the temperature ranges, each component may be separated with high purity through upper and lower portions, and in particular, isopropyl alcohol may not be discharged together to the upper discharge stream supplied to the inert gas removal column **400**.

(45) According to an exemplary embodiment of the present invention, the total number of stages of the second gas purification column **300** may be 25 to 40, 28 to 40, or 28 to 35. In addition, a supply stage of the stream supplied to the second gas purification column **300** may be at 40% to 55%, 43% to 55%, or 45% to 53% of the total number of stages of the second gas purification column **300**.

For example, when the total number of stages of the second gas purification column **300** is 32, the stream supplied to the second gas purification column **300** may be supplied to the 12th to 16th stages of the second gas purification column **300**. As such, by controlling the total number of stages of the second gas purification column **300** and the supply stage of the stream supplied to the second gas purification column **300**, the entire amounts of isopropyl alcohol and unreacted water in the stream supplied to the second gas purification column **300** are separated as a lower discharge stream, so that isopropyl alcohol and water may not be included in the upper discharge stream from the second gas purification column **300** supplied to the inert gas removal column **400**.

(46) According to an exemplary embodiment of the present invention, an upper discharge stream including the unreacted propylene monomer from the supplied stream may be separated in the inert gas removal column **400**, refluxed to the reactor, and reused. Here, no isopropyl alcohol may be present in the upper discharge stream from the inert gas removal column **400**.

(47) According to an exemplary embodiment of the present invention, the upper discharge stream from the inert gas removal column **400** is a gaseous stream and inert gas, for example, ethane, is separated and removed, thereby preventing accumulation of the inert gas in the process to increase the production of isopropyl alcohol without use of a high-purity propylene monomer. Specifically, in the isopropyl alcohol preparation process performed in a gaseous reaction, a part of inert gas may be included in the propylene monomer introduced as a reactant. The inert gas may include, for example, one or more selected from the group consisting of hydrocarbons having 2 or 3 carbon atoms, and as a specific example, the inert gas may include one or more selected from the group consisting of ethane and propane. Here, the gaseous component including the inert gas in the upper discharge stream from the inert gas removal column **400** is purged to separate and remove a part or all of the amount of the inert gas, thereby lowering the content of the inert gas in the stream refluxed to the reactor so that the inert gas is not accumulated in the process.

(48) According to an exemplary embodiment of the present invention, the lower discharge stream from the inert gas removal column **400** is a liquid stream including inert gas, for example, propane, and passes through a third valve **620** to be depressurized and supplied to a gas-liquid separation device **700**. In the gas-liquid separation device **700**, the inert gas may be separated into an upper portion and removed by purging with the gaseous components in the upper discharge stream from the inert gas removal column **400**. In this scenario, no stream discharged to the lower portion of the

gas-liquid separation device **700** may be present because isopropyl alcohol and unreacted water are not present in the lower discharge stream from the inert gas removal column **400**.

(49) According to an exemplary embodiment of the present invention, the operating pressure of the inert gas removal column **400** may be 15 kg/cm.sup.2.Math.g to 25 kg/cm.sup.2.Math.g, 18 kg/cm.sup.2.Math.g to 25 kg/cm.sup.2.Math.g, or 18 kg/cm.sup.2.Math.g to 20 kg/cm.sup.2.Math.g, and the operating temperature thereof may be 30° C. to 70° C., 40° C. to 70° C., or 40° C. to 60° C. By operating the inert gas removal column **400** in these pressure and temperature ranges, the unreacted propylene monomer may be separated with high purity into an upper portion, refluxed to the reactor, and reused.

(50) According to an exemplary embodiment of the present invention, the total number of stages of the inert gas removal column **400** may be 65 to 80, 65 to 78, or 67 to 75. In addition, a supply stage of the stream supplied to the inert gas removal column **400** may be at 50% or less, 1% to 50%, 5% to 45%, or 10% to 45% of the total number of stages of the inert gas removal column **400**. For example, when the total number of stages of the inert gas removal column **400** is 70, the stream supplied to the inert gas removal column **400** may be supplied at the 35th or lower stage of the inert gas removal column **400**. As such, by controlling the total number of stages of the inert gas removal column **400** and the supply stage of the stream supplied to the inert gas removal column **400**, the unreacted propylene monomer may be separated with high purity.

(51) According to an exemplary embodiment of the present invention, a ratio of the total number of stages of the second gas purification column **300** to the inert gas removal column **400** may be 1:1.5 to 1:5, 1:2 to 1:4, or 1:2 to 1:3. As described above, by controlling the ratio of the total number of stages of the second gas purification column **300** to the inert gas removal column **400**, the composition of the upper discharge stream and the lower discharge stream from the second gas purification column **300** may be controlled, and also isopropyl alcohol may be prevented from being included in the upper discharge stream from the inert gas removal column **400**.

(52) According to an exemplary embodiment of the present invention, the stream supplied to the IPA purification section may include isopropyl alcohol, unreacted water, and water used as a solvent, and high-purity isopropyl alcohol from which water is removed in the IPA purification section may be separated. Here, water separated in the IPA purification section may be supplied to the reactor and reused, and in this case, impurities such as a propylene monomer or isopropyl alcohol are not included, so that it may be easy to control a mole ratio of water to the propylene monomer in preparing isopropyl alcohol in the reactor.

(53) According to an exemplary embodiment of the present invention, devices such as a distillation column, a condenser, a reboiler, a valve, a pump, a separator, and a mixer may be further installed and used in the method of preparing isopropyl alcohol if necessary.

(54) Hereinabove, the method of preparing isopropyl alcohol according to the present invention has been described and illustrated in the drawings, but the description and the illustration in the drawings are the description and the illustration of only core constitutions for understanding of the present invention, and in addition to the process and apparatus described above and illustrated in the drawings, the process and the apparatus which are not described and illustrated separately may be appropriately applied and used for carrying out the method of preparing isopropyl alcohol according to the present invention.

(55) Hereinafter, the present invention will be described in more detail by the Examples. However, the following Examples are provided for illustrating the present invention, and it is apparent to a person skilled in the art that various modifications and alterations may be made without departing from the scope and spirit of the present invention and the scope of the present invention is not limited thereto.

EXAMPLES

Example 1

(56) As shown in the process flow diagram of FIG. 1, a reaction product including isopropyl

alcohol (IPA) was prepared, and a process of separating isopropyl alcohol from the reaction product was simulated using Aspen Plus available from Aspen Technology Inc.

(57) Specifically, a feed stream was supplied at a flow rate of 10 ton/hr to a reactor operating at a pressure of 40 kg/cm².Math.g and a temperature of 192° C. to 208° C., a mole ratio of water (H₂O) to a propylene monomer (PP) in the feed stream was controlled to 0.4, and the inert gas included ethane and propane.

(58) A gaseous reaction product stream discharged from the reactor passed through two heat exchangers and was separated into and discharged as a gaseous first reaction product stream and a liquid second reaction product stream, and the first reaction product stream was partly condensed into a liquid phase in a cooler and then supplied to the 20th stage of an absorption column **100**, and the second reaction product stream was supplied to the 20th stage of the absorption column **100**. The total number of stages of the absorption column **100** was 20.

(59) The absorption column **100** was operated at a temperature of 96° C. to 98° C. and a pressure of 31.2 kg/cm².Math.g to 31.6 kg/cm².Math.g, isopropyl alcohol in the reaction product stream was absorbed using water supplied as a solvent in the upper end of the absorption column **100**, and separated into an upper discharge stream including a propylene monomer and a lower discharge stream including water and isopropyl alcohol. Specifically, a part of the upper discharge stream from the absorption column **100** was supplied to a second gas purification column **300**, the rest of the stream was refluxed to the reactor, and the lower discharge stream was depressurized using a first valve **600** and then supplied to a first flash drum **500**. The first flash drum **500** was operated at an operating pressure of 16.8 kg/cm².Math.g and the flow rate of the upper discharge stream was controlled to 0.2 ton/hr.

(60) The lower discharge stream in the first flash drum **500** was depressurized using a second valve **610** and then supplied to the second flash drum **510**, and the upper discharge stream from the second flash drum **510** was supplied to the first gas purification column **200** and the lower discharge stream from the second flash drum was supplied to an IPA purification section. The second flash drum **510** was operated at an operating pressure of 0.2 kg/cm².Math.g and the flow rate of the upper discharge stream was controlled to 0.1 ton/hr.

(61) The lower discharge stream from the first gas purification column **200** was supplied to the IPA purification section. The total number of stages of the first gas purification column **200** was 4, and the operating temperature was controlled to 57° C. to 64° C. and the operating pressure was controlled to 0.15 kg/cm².Math.g to 0.2 kg/cm².Math.g.

(62) A part of the upper discharge stream from the absorption column **100**, the upper discharge stream from the gas purification column **200**, and the upper discharge stream from the first flash drum **500** formed a mixed stream and was supplied to the 16th stage of the second gas purification column **300**. The total number of stages of the second gas purification column **300** was 32, and the operating temperature was controlled to 41° C. to 160° C. and the operating pressure was controlled to 16.50 kg/cm².Math.g to 16.52 kg/cm².Math.g. An upper discharge stream from the second gas purification column **300** was supplied to the 26th stage of the inert gas removal column **400** and a lower discharge stream from the second gas purification column was supplied to the IPA purification section. The total number of stages of the inert gas removal column **400** was 70.

(63) The inert gas removal column **400** was operated at a temperature of 47° C. to 55° C. and at a pressure of 19.10 kg/cm².Math.g to 19.59 kg/cm².Math.g, and the unreacted propylene monomer was separated as an upper discharge stream, refluxed to the reactor, and reused.

(64) (1) The IPA content in the upper discharge stream from the inert gas removal column **400** refluxed to the reactor, (2) the PP content in the upper discharge stream from the inert gas removal column **400** refluxed to the reactor, and (3) the PP content in the stream supplied to the IPA purification section are shown in Table 1.

Example 2

(65) As shown in the process flow diagram of FIG. 2, a reaction product including isopropyl alcohol (IPA) was prepared, and a process of separating isopropyl alcohol from the reaction product was simulated using Aspen Plus available from Aspen Technology Inc.

(66) Specifically, a feed stream was supplied at a flow rate of 10 ton/hr to a reactor operating at a pressure of 40 kg/cm².Math.g and a temperature of 192° C. to 208° C., a mole ratio of water (H₂O) to a propylene monomer (PP) in the feed stream was controlled to 0.4, and the inert gas included ethane and propane.

(67) A gaseous reaction product stream discharged from the reactor passed through two heat exchangers and was separated into and discharged as a gaseous first reaction product stream and a liquid second reaction product stream, and the first reaction product stream was partly condensed into a liquid phase in a cooler and then supplied to the 20th stage of an absorption column **100**, and the second reaction product stream was supplied to the 20th stage of the absorption column **100**. The total number of stages of the absorption column **100** was 20.

(68) The absorption column **100** was operated at a temperature of 96° C. to 98° C. and a pressure of 31.2 kg/cm².Math.g to 31.6 kg/cm².Math.g, isopropyl alcohol in the reaction product stream was absorbed using water supplied as a solvent in the upper end of the absorption column **100**, and separated into an upper discharge stream including a propylene monomer and a lower discharge stream including water and isopropyl alcohol. Specifically, a part of the upper discharge stream from the absorption column **100** was supplied to the second gas purification column **300**, the rest of the stream was refluxed to the reactor, and the lower discharge stream was depressurized using the first valve **600** and then supplied to a first flash drum **500**. The first flash drum **500** was operated at an operating pressure of 16.8 kg/cm².Math.g and the flow rate of the upper discharge stream was controlled to 0.2 ton/hr.

(69) The lower discharge stream in the first flash drum **500** was depressurized using a second valve **610** and then supplied to the second flash drum **510**, and the upper discharge stream from the second flash drum **510** was supplied to the first gas purification column **200** and the lower discharge stream from the second flash drum was supplied to an IPA purification section. The second flash drum **510** was operated at an operating pressure of 0.2 kg/cm².Math.g and the flow rate of the upper discharge stream was controlled to 0.1 ton/hr.

(70) The lower discharge stream from the first gas purification column **200** was supplied to the IPA purification section. The total number of stages of the first gas purification column **200** was 4, and the operating temperature was controlled to 57° C. to 64° C. and the operating pressure was controlled to 0.15 kg/cm².Math.g to 0.2 kg/cm².Math.g.

(71) A part of the upper discharge stream from the absorption column **100**, the upper discharge stream from the gas purification column **200**, and the upper discharge stream from the first flash drum **500** formed a mixed stream and was supplied to the 16th stage of the second gas purification column **300**. The total number of stages of the second gas purification column **300** was 32, and the operating temperature was controlled to 41° C. to 160° C. and the operating pressure was controlled to 16.50 kg/cm².Math.g to 16.52 kg/cm².Math.g. An upper discharge stream from the second gas purification column **300** was supplied to the 26th stage of the inert gas removal column **400** and a lower discharge stream from the second gas purification column was supplied to the IPA purification section. The total number of stages of the inert gas removal column **400** was 70.

(72) The inert gas removal column **400** was operated at a temperature of 47° C. to 55° C. and at a pressure of 19.10 kg/cm².Math.g to 19.59 kg/cm².Math.g, and an upper discharge stream of liquid components including the unreacted propylene monomer was refluxed to the reactor and reused. In addition, the gaseous components including inert gas in the upper discharge stream from the inert gas removal column **400** was purged to be removed, a lower discharge stream from the inert gas removal column passed through a third valve **620** to be depressurized, and was supplied to a gas-liquid separation device **700**, and inert gas was separated into an upper portion in the gas-

liquid separation device **700** and removed.

(73) (1) The IPA content in the upper discharge stream from the inert gas removal column **400** refluxed to the reactor, (2) the PP content in the upper discharge stream from the inert gas removal column **400** refluxed to the reactor, and (3) the PP content in the stream supplied to the IPA purification section are shown in Table 1.

Example 3

(74) Isopropyl alcohol was prepared in the same manner as in Example 2, except that the upper discharge stream from the second gas purification column **300** was supplied to the 7th stage of the inert gas removal column **400**.

(75) (1) The IPA content in the upper discharge stream from the inert gas removal column **400** refluxed to the reactor, (2) the PP content in the upper discharge stream from the inert gas removal column **400** refluxed to the reactor, and (3) the PP content in the stream supplied to the IPA purification section are shown in Table 1.

Example 4

(76) Isopropyl alcohol was prepared in the same manner as in Example 2, except that the mixed stream was supplied to the 4th stage of the second gas purification column **300**.

(77) (1) The IPA content in the upper discharge stream from the inert gas removal column **400** refluxed to the reactor, (2) the PP content in the upper discharge stream from the inert gas removal column **400** refluxed to the reactor, and (3) the PP content in the stream supplied to the IPA purification section are shown in Table 1.

Example 5

(78) Isopropyl alcohol was prepared in the same manner as in Example 2, except that the mixed stream was supplied to the 30th stage of the second gas purification column **300**.

(79) (1) The IPA content in the upper discharge stream from the inert gas removal column **400** refluxed to the reactor, (2) the PP content in the upper discharge stream from the inert gas removal column **400** refluxed to the reactor, and (3) the PP content in the stream supplied to the IPA purification section are shown in Table 1.

Example 6

(80) Isopropyl alcohol was prepared in the same manner as in Example 2, except that the upper discharge stream from the second gas purification column **300** was supplied to the 38th stage of the inert gas removal column **400**.

(81) (1) The IPA content in the upper discharge stream from the inert gas removal column **400** refluxed to the reactor, (2) the PP content in the upper discharge stream from the inert gas removal column **400** refluxed to the reactor, and (3) the PP content in the stream supplied to the IPA purification section are shown in Table 1.

Example 7

(82) Isopropyl alcohol was prepared in the same manner as in Example 2, except that the mixed stream was supplied to the 30th stage of the second gas purification column **300** and the upper discharge stream from the second gas purification column **300** was supplied to the 38th stage of the inert gas removal column **400**.

(83) (1) The IPA content in the upper discharge stream from the inert gas removal column **400** refluxed to the reactor, (2) the PP content in the upper discharge stream from the inert gas removal column **400** refluxed to the reactor, and (3) the PP content in the stream supplied to the IPA purification section are shown in Table 1.

COMPARATIVE EXAMPLES

Comparative Example 1

(84) As shown in the process flow diagram of FIG. 3, a reaction product including isopropyl alcohol (IPA) was prepared, and a process of separating isopropyl alcohol from the reaction product was simulated using Aspen Plus available from Aspen Technology Inc.

(85) Specifically, a feed stream was supplied at a flow rate of 10 ton/hr to a reactor operating at a

pressure of 40 kg/cm.sup.2.Math.g and a temperature of 192° C. to 208° C., a mole ratio of water (H.sub.2O) to a propylene monomer (PP) in the feed stream was controlled to 0.4, and the inert gas included ethane and propane.

(86) The reaction product stream discharged from the reactor was supplied to the 20th stage of the absorption column **100**. The total number of stages of the absorption column **100** was 20. The absorption column **100** was operated at a temperature of 96° C. to 98° C. and a pressure of 31.2 kg/cm.sup.2.Math.g to 31.6 kg/cm.sup.2.Math.g, isopropyl alcohol in the reaction product stream was absorbed using water supplied as a solvent in the upper end of the absorption column **100**, and separated into an upper discharge stream including a propylene monomer and a lower discharge stream including water and isopropyl alcohol. Specifically, a part of the upper discharge stream from the absorption column **100** was supplied to the gas purification column **310**, the rest of the stream was refluxed to the reactor, and the lower discharge stream was supplied to the flash drum **520**.

(87) In the gas purification column **310**, the upper discharge stream including the unreacted propylene monomer was refluxed to the reactor, and the lower discharge stream was supplied to the IPA purification section. At this time, the total number of stages of the gas purification column **310** was 32, and the operating temperature was controlled to 41° C. to 160° C. and the operating pressure was controlled to 16.50 kg/cm.sup.2.Math.g to 16.52 kg/cm.sup.2.Math.g.

(88) In the flash drum **520**, the upper discharge stream including the gaseous unreacted propylene monomer was refluxed to the reactor, and the lower discharge stream was supplied to the IPA purification section. At this time, the flash drum was operated at an operating pressure of 16.8 kg/cm.sup.2.Math.g and the flow rate of the upper discharge stream was controlled to 0.2 ton/hr.

(89) (1) The IPA content in the upper discharge stream from the gas purification column **310** and the upper discharge stream from the flash drum **520** refluxed to the reactor, (2) the PP content in the upper discharge stream from the gas purification column **310** and the upper discharge stream from the flash drum **520** refluxed to the reactor, and (3) the PP content in the stream supplied to the IPA purification section are shown in Table 1.

(90) TABLE-US-00001 TABLE 1 Comparative Example Example 1 2 3 4 5 6 7 1 (1) (wt %) 0 0 0 0 0 0 0 3.5 (2) (wt %) 94.9 94.9 95.1 94.9 94.8 94.8 94.7 83.2 (3) (wt %) 0.006 0.006 0.006 0.009 0.014 0.006 0.014 0.4

(91) As shown by the results of Examples 1 to 7 in Table 1, the content of isopropyl alcohol in the stream refluxed to the reactor was 0 wt %, the content of the propylene monomer was as high as 94.7 wt % or more, and the content of the propylene monomer in the stream supplied to the IPA purification section was as low as 0.014 wt % or less.

(92) In addition, in Examples 2 to 7, the inert gas in the upper discharge stream from the inert gas removal column **400** was removed and then the stream was refluxed to the reactor and reused, thereby preventing accumulation of the inert gas in the process, and thus, the production of isopropyl alcohol was increased without the use of a high-purity propylene monomer.

(93) Separation efficiency was somewhat lowered in Examples 4 to 7, in which any one or more of a supply stage at which the mixed stream was supplied to the second gas purification column **300** and a supply stage at which the stream was supplied to the inert gas removal column **400** were out of the range according to the present invention, due to lack of a rectifying section to separate a high-boiling point material into a lower portion or lack of a stripping section to separate a low-boiling point material into an upper portion.

(94) In comparison, when the unreacted propylene monomer and isopropyl alcohol were separated from the reaction product stream using the absorption column **100**, the flash drum **520**, and the gas purification column **310**, as described in Comparative Example 1, the content of isopropyl alcohol was as high as 3.5 wt % and the content of the propylene monomer was as low as 83.2 wt % in the upper discharge stream from the flash drum **520** and the upper discharge stream from the gas purification column **310** refluxed to the reactor. Thus, a reverse reaction was caused in the reactor,

which reduced the production of isopropyl alcohol. In addition, the content of the propylene monomer in the stream supplied to the IPA purification section was as high as 0.4 wt %. Thus, an amount of the propylene monomer which was refluxed to the reactor and lost without being reused was increased.

Claims

1. A method of preparing isopropyl alcohol, the method comprising: reacting a propylene monomer and water in a reactor to produce a reaction product stream; supplying the reaction product stream to an absorption column; supplying a lower discharge stream including isopropyl alcohol from the absorption column to a first gas purification column; supplying a part of an upper discharge stream including an unreacted propylene monomer from the absorption column to a second gas purification column; refluxing a remaining part of the upper discharge stream from the absorption column to the reactor; supplying an upper discharge stream including the unreacted propylene monomer from the first gas purification column to the second gas purification column; supplying a lower discharge stream including isopropyl alcohol from the first gas purification column to an isopropyl alcohol (IPA) purification section; supplying an upper discharge stream including the unreacted propylene monomer from the second gas purification column to an inert gas removal column; supplying a lower discharge stream including isopropyl alcohol from the second gas purification column to the IPA purification section; and refluxing an upper discharge stream including the unreacted propylene monomer from the inert gas removal column to the reactor.
2. The method of preparing isopropyl alcohol of claim 1, further comprising: supplying the lower discharge stream from the absorption column to a first flash drum; supplying an upper discharge stream from the first flash drum to the second gas purification column; supplying a lower discharge stream from the first flash drum to a second flash drum; supplying an upper discharge stream from the second flash drum to the first gas purification column; and supplying a lower discharge stream from the second flash drum to the IPA purification section.
3. The method of preparing isopropyl alcohol of claim 2, further comprising: passing the lower discharge stream from the absorption column through a first valve to produce a depressurized lower discharge stream from the absorption column; supplying the depressurized lower discharge stream from the absorption column to the first flash drum; passing the lower discharge stream from the first flash drum through a second valve to produce a depressurized lower discharge stream from the first flash drum; and supplying the depressurized lower discharge stream from the first flash drum to the second flash drum.
4. The method of preparing isopropyl alcohol of claim 2, wherein a flow rate ratio between a flow rate of the upper discharge stream from the first flash drum and a flow rate of the upper discharge stream from the second flash drum is 1:0.2 to 1:0.8.
5. The method of preparing isopropyl alcohol of claim 1, further comprising: separating inert gas from the upper discharge stream and the lower discharge stream of the inert gas removal column.
6. The method of preparing isopropyl alcohol of claim 1, further comprising: separating the reaction product stream into a first reaction product stream and a second reaction product stream by passing the reaction product stream through one or more heat exchangers; and supplying the first reaction product stream and the second reaction product stream to the absorption column.
7. The method of preparing isopropyl alcohol of claim 1, wherein a total number of stages of the second gas purification column is 25 to 40, and wherein a supply stage where the part of the upper discharge stream from the absorption column and the upper discharge stream from the first gas purification column is supplied to the second gas purification column is a stage at 40% to 55% of the total number of stages of the second gas purification column.
8. The method of preparing isopropyl alcohol of claim 1, wherein an operating pressure of the second gas purification column is 12 kg/cm².Math.g to 20 kg/cm².Math.g, and an

operating temperature thereof is 30° C. to 170° C.

9. The method of preparing isopropyl alcohol of claim 1, wherein a total number of stages of the inert gas removal column is 65 to 80, and wherein a supply stage where the upper discharge stream from the second gas purification column is supplied to the inert gas removal column is a stage at 50% or lower of the total number of stages of the inert gas removal column.

10. The method of preparing isopropyl alcohol of claim 1, wherein an operating pressure of the inert gas removal column is 15 kg/cm.sup.2.Math.g to 25 kg/cm.sup.2.Math.g.

11. The method of preparing isopropyl alcohol of claim 1, wherein a ratio of a total number of stages of the second gas purification column and a total number of stages of the inert gas removal column is 1:1.5 to 1:5.

12. The method of preparing isopropyl alcohol of claim 1, wherein an operating temperature of the second gas purification column is 30° C. to 170° C.

13. The method of preparing isopropyl alcohol of claim 1, wherein an operating temperature of the inert gas removal column is 30° C. to 70° C.
